

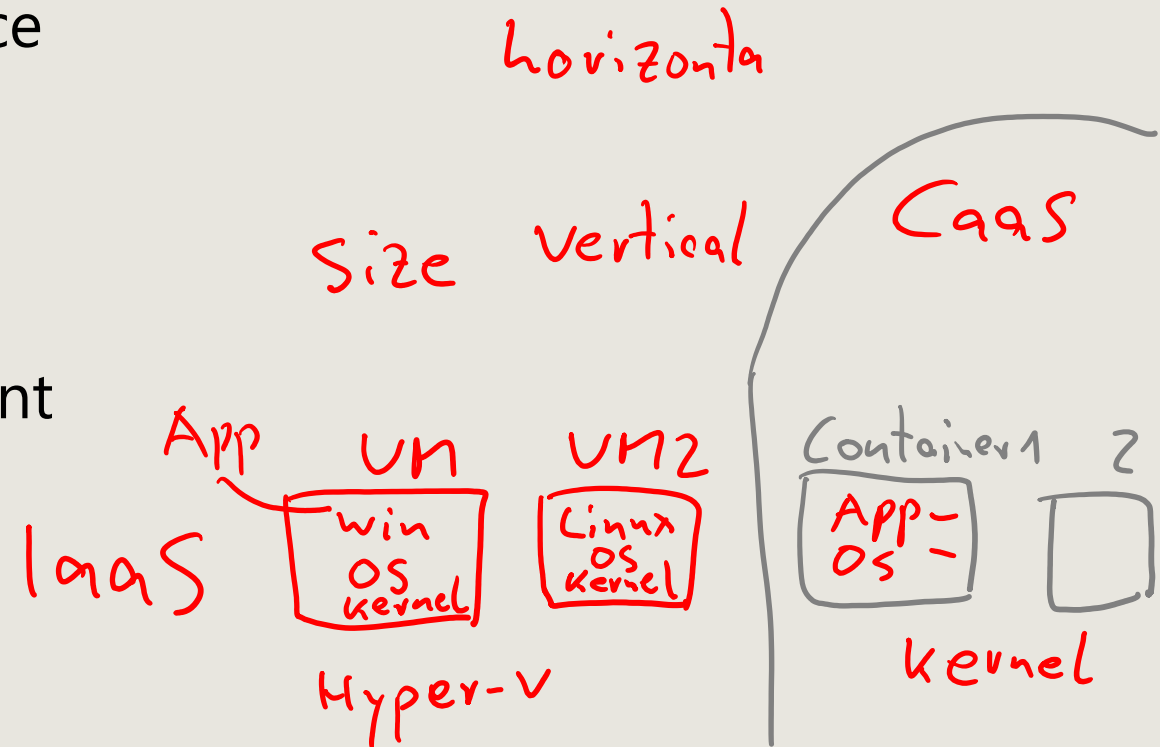


AZ-104

Administer Azure Virtual Machines

AZ-104 Agenda

- 01: Administer Identity
- 02: Administer Governance and Compliance
- 03: Administer Azure Resources
- 04: Administer Virtual Networking
- 05: Administer Intersite Connectivity
- 06: Administer Network Traffic Management
- 07: Administer Azure Storage
- 08: Administer Azure Virtual Machines
- 09: Administer PaaS Compute Options
- 10: Administer Data Protection
- 11: Administer Monitoring



Learning Objectives - Administer Azure Virtual Machines

- Introduction to Azure Virtual Machines
- Configure Virtual Machine Availability
- Lab 08 – Manage Virtual Machines

Introduction to Azure Virtual Machines



Learning Objectives – Introduction to Azure Virtual Machines

- Review Cloud Services Responsibilities
- Plan Virtual Machines
- Determine Virtual Machine Sizing
- Determine Virtual Machine Storage
- Connect to Virtual Machines
- Demonstration - Create and connect to virtual machines
- Learning Recap

Exam objectives

Create and configure virtual machines (VMs)

- Create a VM
- Move VMs between resource groups
- Manage VM sizes
- Manage VM disks

Configure secure access to virtual networks

- Implement Azure Bastion

Review Cloud Services Responsibilities

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Microsoft	Shared	Customer	Customer
	Network controls	Microsoft	Shared	Customer	Customer
	Operating system	Microsoft	Microsoft	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

 Microsoft	 Customer	 Shared
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Plan Virtual Machines



- Start with the network
- Name the virtual machine
- Choose a location
 - Each region has different hardware and service capabilities
 - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing

* Same Region



Determine Virtual Machine Sizing

Type	Description
General purpose	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

Determine Virtual Machine Storage

Each Azure VM has two or more disks:

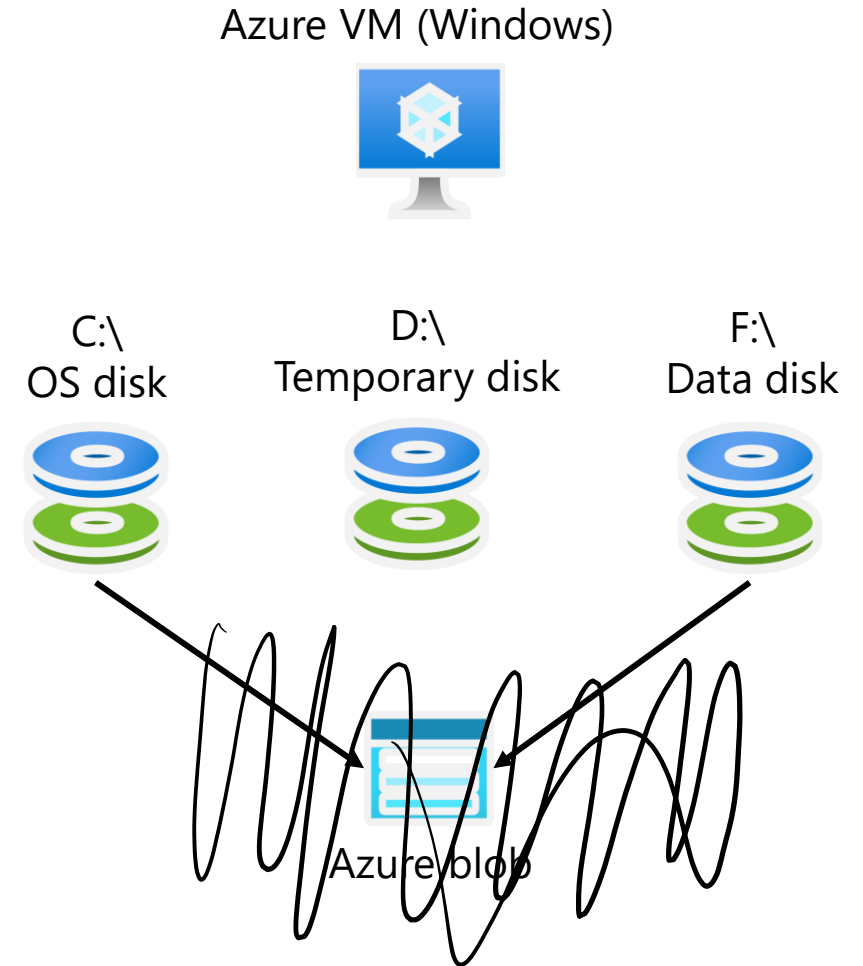
- OS disk
- Temporary disk (not all SKUs have one, content can be lost)

- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

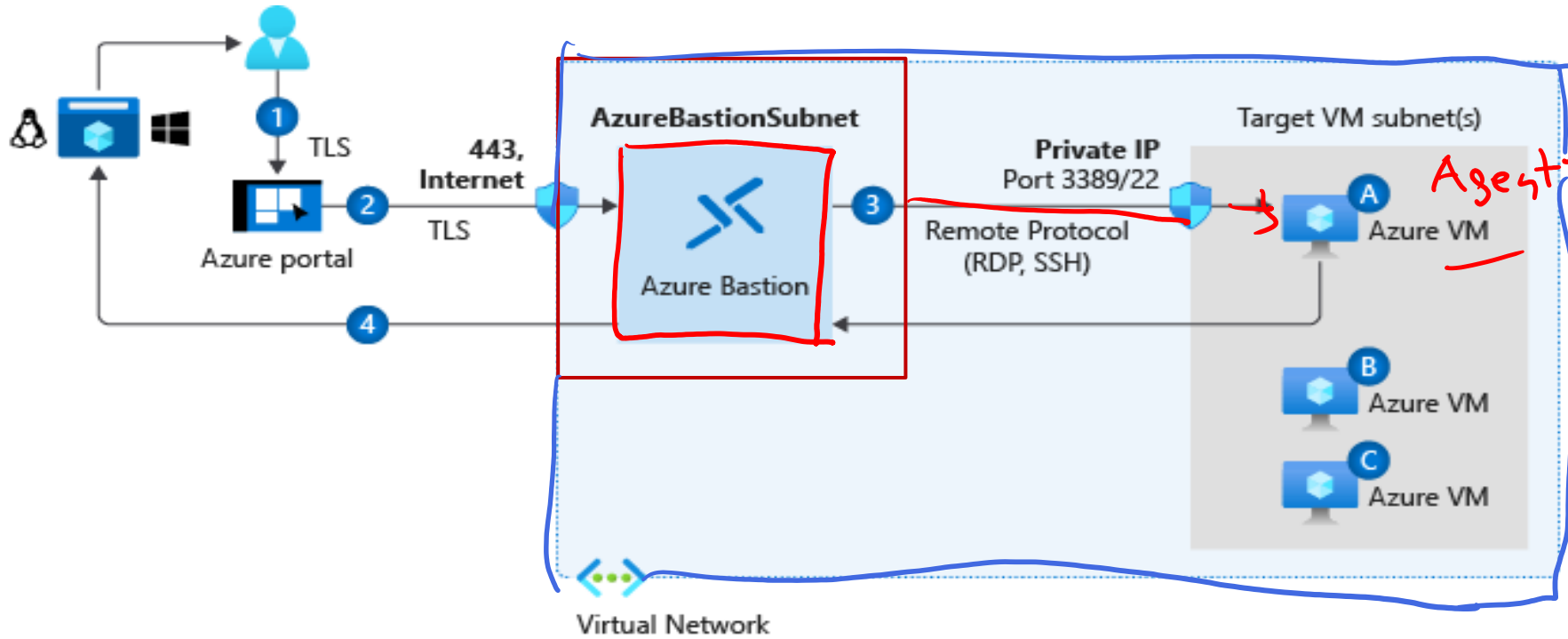
- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks



Connect to Virtual Machines

Portal
Run Command



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Configure Virtual Machine Availability



Configure Azure Virtual Machine Availability Introduction

- Plan for Maintenance and Downtime
- Setup Availability Sets
- Review Update and Fault Domains
- Review Availability Zones
- Compare Vertical to Horizontal Scaling
- Create and Configure Scaling
- Demonstration – Virtual Machine Scaling
- Bring Azure innovation to your hybrid environments with Azure Arc (optional)
- Learning Recap

Exam objectives

Create and configure virtual machines

- Deploy virtual machines to availability zones and availability sets
- Deploy and configure an Azure Virtual Machine Scale Sets

Setup Availability Sets

Instance details

Name * ⓘ

avset01 ✓

Region * ⓘ

(US) East US ✓

Fault domains ⓘ

Rack

2

Update domains ⓘ

5

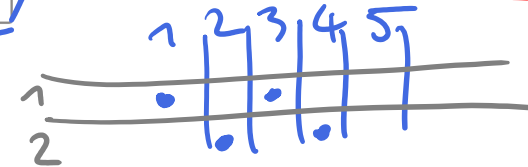
Use managed disks ⓘ

No (Classic)

Yes (Aligned)

Avail Zone
= Data Center

Two or more instances in
Availability Sets = 99.95% SLA



Configure multiple
Virtual Machines in
an Availability Set

Configure each
application tier
into separate
Availability Sets

Combine a Load
Balancer with
Availability Sets

Use managed disks
with the Virtual
Machines

Review Availability Zones

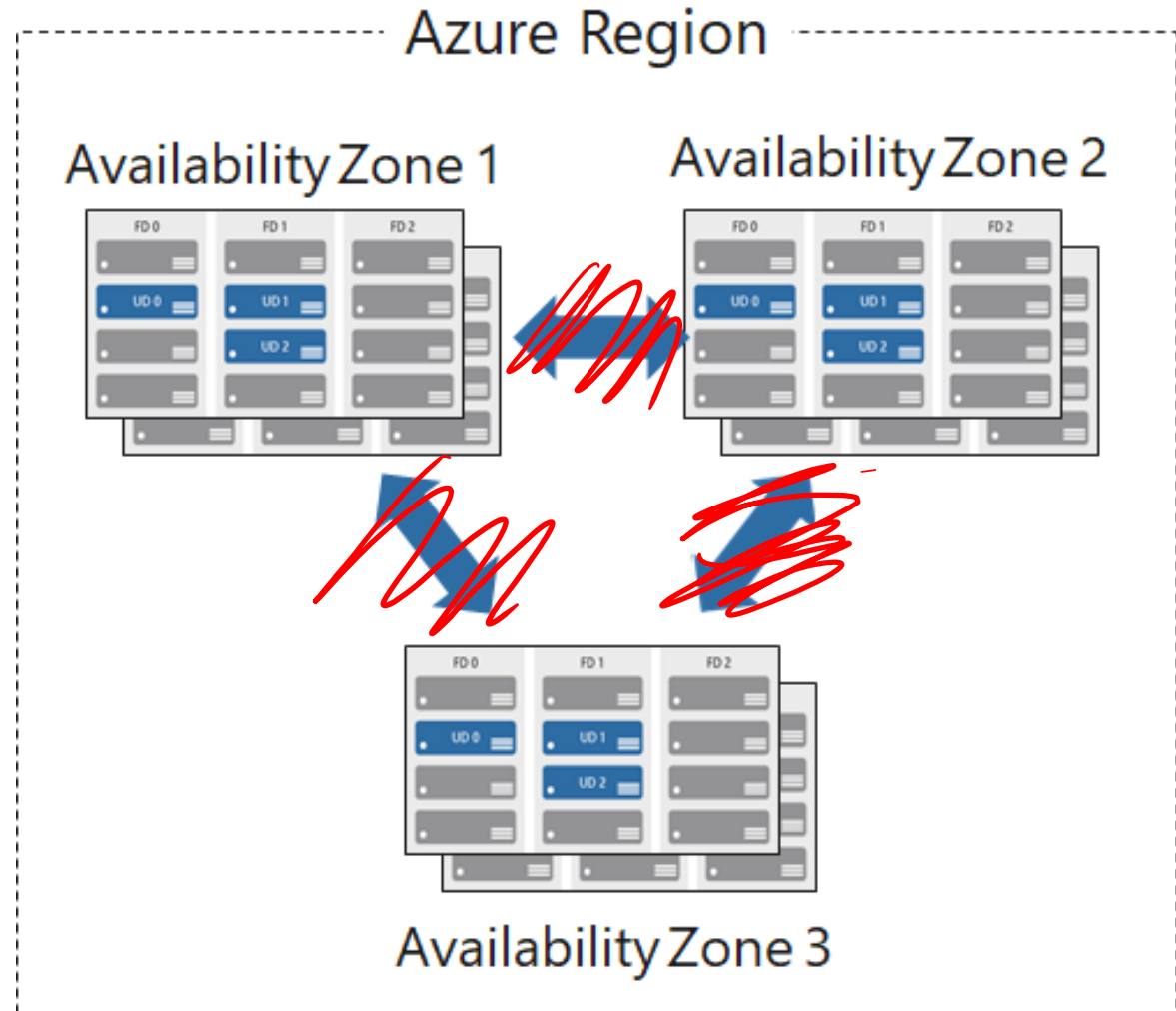
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

Combines update and fault domains

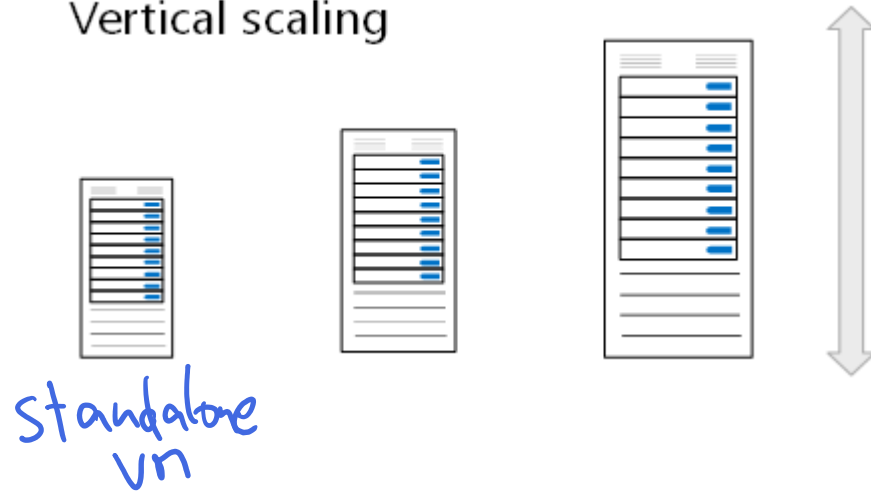
Provides 99.99% SLA



Compare Vertical to Horizontal Scaling

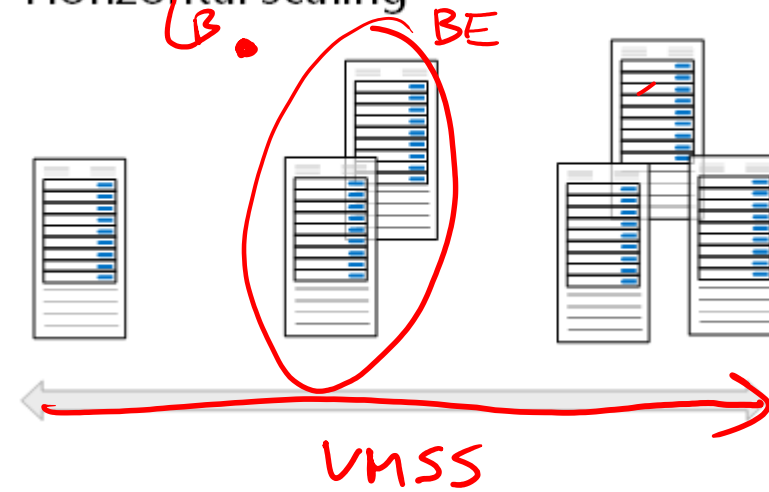
VMSS — uniform flex

Vertical scaling



Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

Horizontal scaling



Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters

Add a scaling condition

×

Scale mode

☐ Manually update the capacity: Scaling based on a CPU metric, on any schedule

☒ Autoscaling: Scaling based on a CPU metric, on any schedule

Default instance count * ⓘ

Instance limit

Minimum * ⓘ

Maximum * ⓘ

Scale out

CPU threshold greater than * ⓘ

Increase instance count by * ⓘ

Scale in

CPU threshold less than * ⓘ

Decrease instance count by * ⓘ

Query duration

Minutes * ⓘ

60

The engine will query CPU usage for the past 60 minutes before executing the scaling to avoid reacting to transient spikes.

Schedule

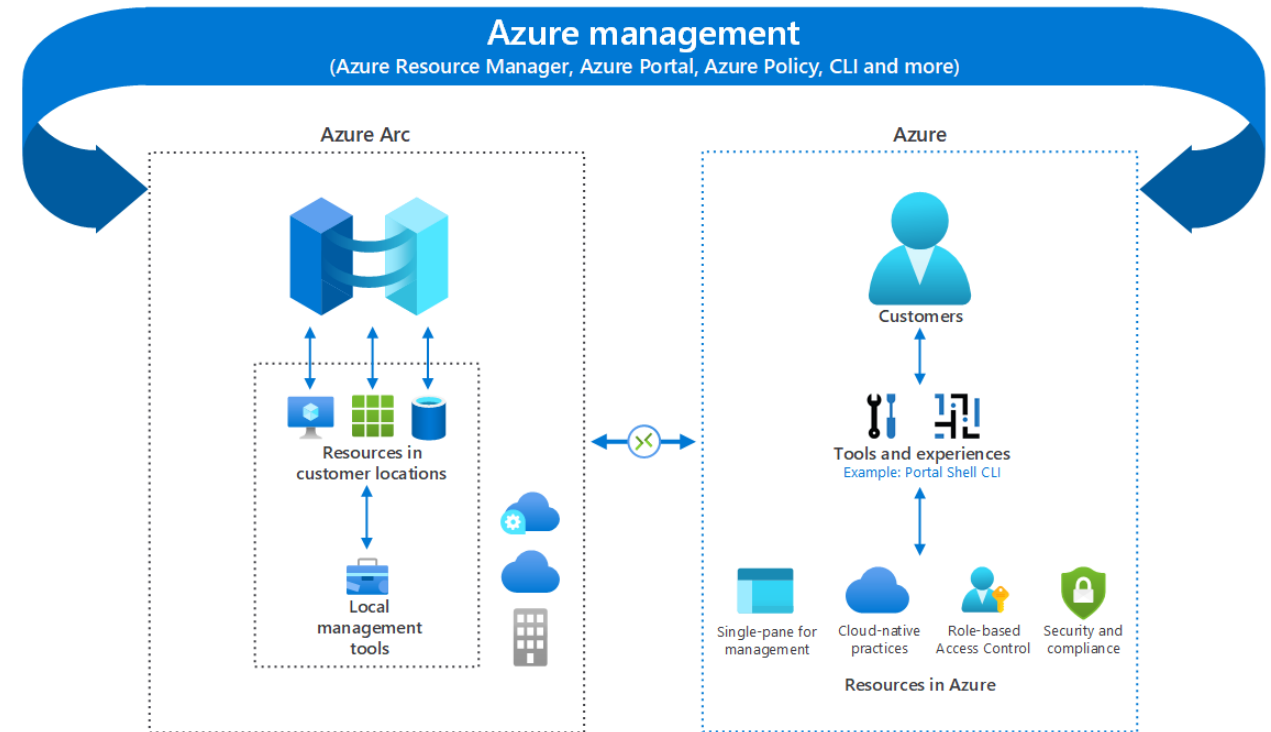
Schedule type *

☒ Specify start/end dates

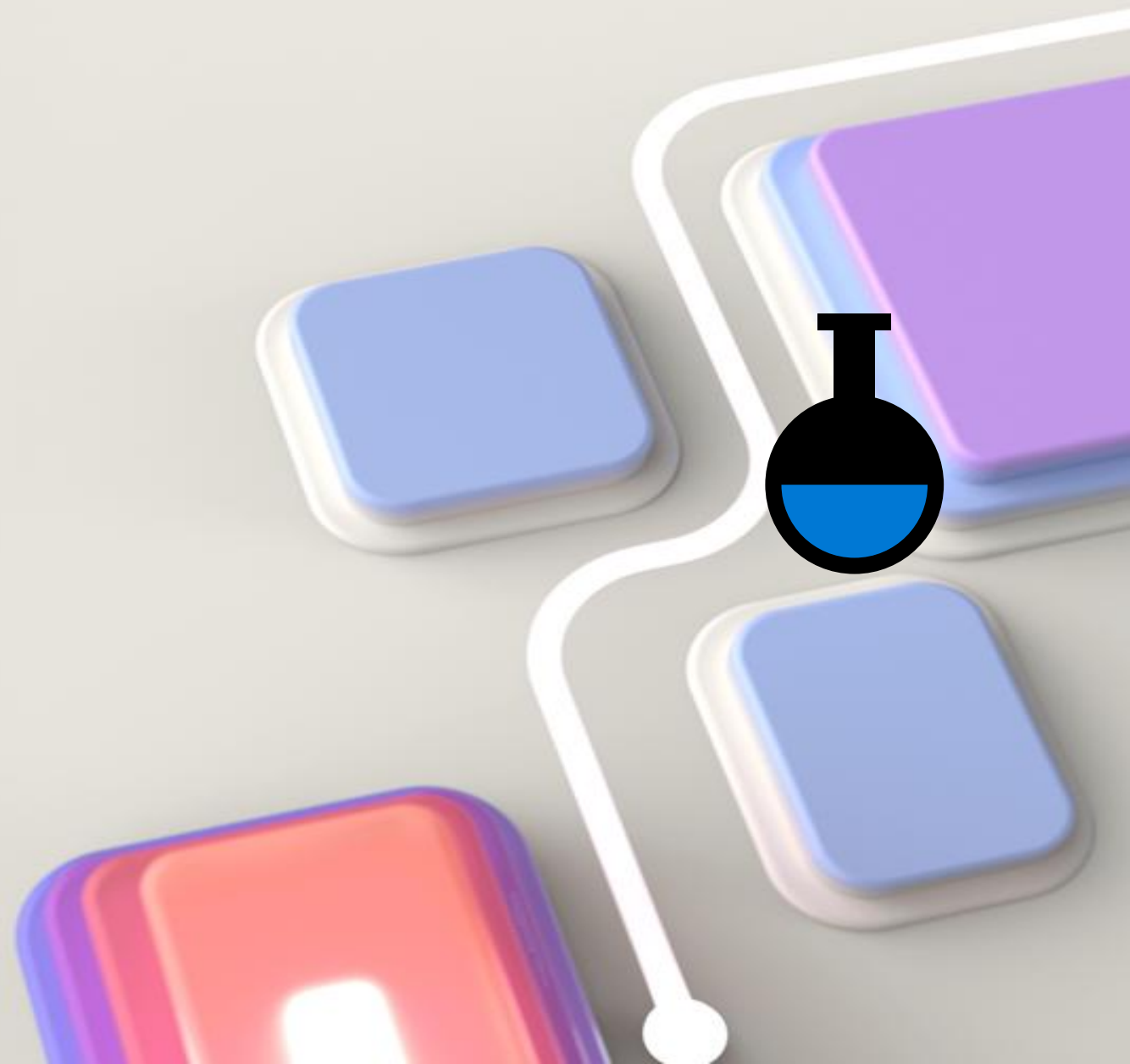
☐ Repeat specific days

Bring Azure innovation to your hybrid environments with Azure Arc

- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment

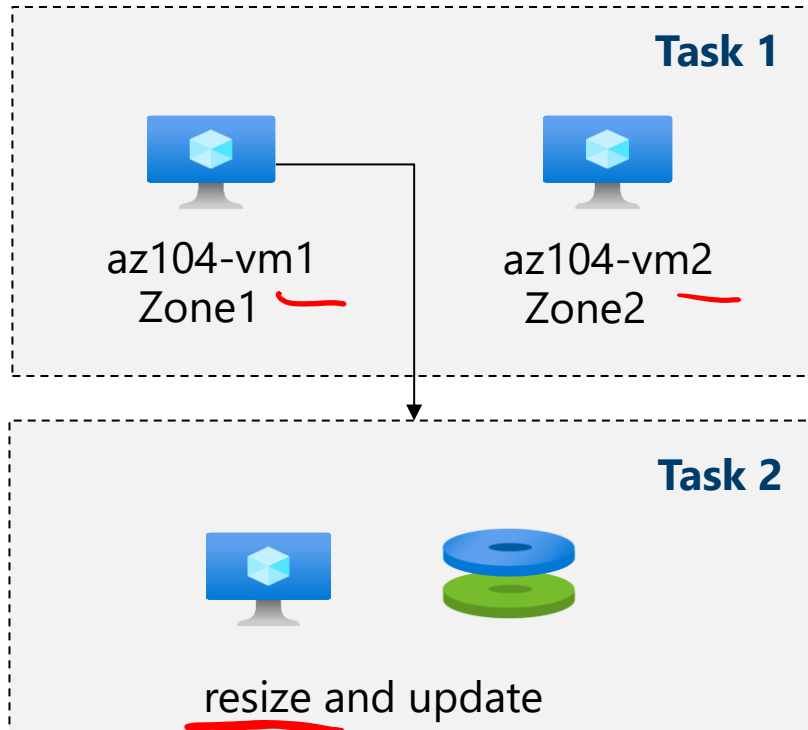


Lab – Manage Virtual Machines

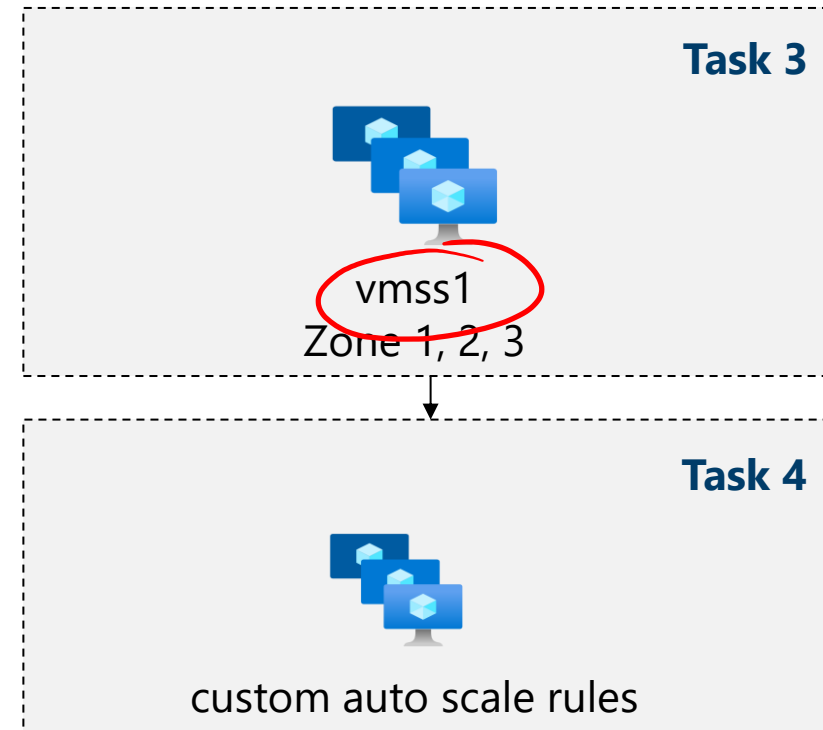


Lab 08 – Architecture diagram

Azure Virtual Machines



Virtual Machine Scale Sets



Task 5: Create a virtual machine using Azure PowerShell (option 1)

Task 6: Create a virtual machine using the CLI (option 2)

End of presentation



